

Answers

Q1. Coin tossed 10 times: H H T H H H T H H T. Estimate probability of Heads using MLE.

Formula:

$$L(p) = p^k(1-p)^{(n-k)}$$

Log-Likelihood:

$$\log L = k \log(p) + (n-k) \log(1-p)$$

MLE:

$$\hat{p} = k/n$$

Here $n=10$, Heads $k=7$.

$$\hat{p} = 7/10 = 0.7$$

Explanation:

MLE estimates the parameter that maximizes the likelihood of observing the given data. Since 7 heads were observed in 10 tosses, the estimated probability of getting heads is 0.7 (70%). This value best explains the observed outcomes under a Bernoulli model.

Result: Estimated probability of Heads = 0.7.

Q2. Out of 50 students, 42 passed. Find MLE of pass probability.

Formula:

$$\hat{p} = k/n$$

Here $n=50$, $k=42$.

$$\hat{p} = 42/50 = 0.84$$

Explanation:

Assuming each student either passes or fails independently, the data follows a Bernoulli distribution. The maximum likelihood estimate equals the sample proportion. Therefore the estimated probability that a student passes is 0.84 or 84%.

Result: Pass probability = 0.84.

Q3. In 100 bulbs, 8 are defective. Find MLE of defect probability.

Formula:

$$\hat{p} = k/n$$

Here $n=100$, $k=8$.

$$\hat{p} = 8/100 = 0.08$$

Explanation:

The defect status of each bulb follows a Bernoulli distribution. MLE estimates the defect probability using the observed proportion of defective bulbs. Hence the probability of a defective bulb is estimated as 0.08 or 8%.

Result: Defect probability = 0.08.

Q4. A biased die is rolled 60 times. Six appears 18 times. Estimate probability using MLE.

Formula:

$$\hat{p} = k/n$$

Here $n=60$, $k=18$.

$$\hat{p} = 18/60 = 0.30$$

Explanation:

Each roll is treated as an independent Bernoulli trial where success is obtaining the face 6. The MLE estimate equals the observed frequency of success. Thus the estimated probability of rolling a 6 is 0.30.

Result: Probability of rolling 6 = 0.30.

Q5. Input $x=4$, Weight $w=2$, Actual output=10, $y=w \times x$. Find predicted output, error and updated weight.

Prediction Formula:

$$y = w \times x$$

Predicted output:

$$y = 2 \times 4 = 8$$

Error:

$$\begin{aligned}\text{Error} &= \text{Actual} - \text{Predicted} \\ &= 10 - 8 = 2\end{aligned}$$

Weight Update Formula:

$$w_{\text{new}} = w_{\text{old}} - \eta \times \text{Gradient}$$

Given:

$$\eta = 0.01$$

$$\text{Gradient} = 4$$

$$\begin{aligned}w_{\text{new}} &= 2 - (0.01 \times 4) \\ &= 2 - 0.04 \\ &= 1.96\end{aligned}$$

Explanation:

The neuron predicts an output of 8 while the expected output is 10, giving an error of 2. Gradient descent updates the weight by moving opposite to the gradient to reduce future error.

Result:

$$\text{Predicted Output} = 8$$

$$\text{Error} = 2$$

$$\text{Updated Weight} = 1.96.$$